ÖZYEĞİN UNIVERSITY	Course Syllabus **EE 302 AI-based Digital Signal Processing** SPRING 2025-26

INSTRUCTOR	
Name	*Cenk Demiroğlu*
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Phone Number	*0216-564-9384*
Office Number	*AB1.315*
Office Hours	*TBD*
Research/Teaching Assistants	*TBD*

COURSE ORGANIZATION	
Interaction Methods	-
Technology Used by the Instructor	Python, Pytorch, Matlab, LMS, VS Code, Google colab extension
Technology Requirements for Students	Laptop, Google colab account, Huggingface account

COURSE MATERIALS	
Required Books	-
Recommended Books	-
Other Materials	*Deeplearning.ai video lectures, pytorch documentation, Huggingface documentation/tutorials*
Accessing Course Materials	-

WEEKLY SUBJECTS (TENTATIVE)		
Week	**Date**	**Subject**
1	9-13.02.2026	Course Introduction and discrete time systems
2	16-20.02.2026	Convolution and z-transform
3	23-27.02.2026	The z-transform
4	02-06.03.2026	Transform analysis of linear time-invariant systems
5	09-13.03.2026	The discrete Fourier transform
6	16-20.03.2026	Fourier analysis of signals using DFT
7	23-27.03.2026	Linear adaptive filters
8	30.03-03.04.2026	Introduction to pytorch
9	06-10.04.2026	Logistic Regression Model and Gradient-Descent
10	13-17.04.2026 ***(Withdrawal Week)***	Regression with Feedforward neural networks
11	20-24.04.2026	Hyperparameter tuning
12	27.04-01.05.2026	1d Convolutional neural networks – Signal denoising
13	04-08.05.2026	2d Convolutional neural networks – İmage denoising
14	11-15.05.2026	Recurrent neural networks - Predictive modelling
15	18-22.05.2026	Recurrent neural networks – Predictive modelling

ASSESSMENT METHODS, WEIGHTS AND RULES			
Type	**Weight**	**Implementation Rule**	**Makeup Rule**
Final Exam	*50*	*It will be on a PC with open-notes and open-codes. Exam will be mostly based on coding questions. Only access to google colab and local course material will be allowed.*	BÜT exam will serve as the makeup for the final exam.
Midterm Exam	*30*	*It will be on a PC with open-notes and open-codes. Exam will be mostly based on coding questions. Only access to google colab and local course material will be allowed.*	*Only valid excuses with an official report are accepted to qualify for a midterm makeup. At most one makeup will be given in the course due to health reports.*
Homework	*0*	*There will be homeworks for practice but they will not be graded.*	
Project	*20*	*There will be a term project, which should be on an algorithm that we do not cover in class. Students should pick their topics by week-7, get an approval from the instructor, and present their work at the end of the semester. Half of the grade will be based on presentation and in-depth understanding of the topic. The other half is on implementation of the idea.*	
Total	**100%**		

METHODS FOR TRACKING ATTENDANCE & IMPACT OF ATTENDANCE ON SUCCESS			
Method	**Application Details**	**Required Tools and Make-up Method**	**Impact of Attendance of Course Grade**
QR Code (MyOzU)	During class, a QR code will be displayed on the screen. Scanning this code with the MyOzU app on your mobile device will be accepted as proof of attendance.	You must have a smartphone with the MyOzU app installed. If you do not have such a device, inform the instructor during class and sign the attendance sheet.	Out of 28 sessions, at least 14 must be attended. Each additional attendance beyond 14 adds 0.5 bonus points (out of 100) to the final exam grade.

ASSESSMENT(S) BEFORE THE WITHDRAWAL DEADLINE		
Type of Assessment(s)	**Weight(s)**	**Date by When the Score of the Assessment is to be Announced**
Midterm	30	12 April 2026.

DETERMINING LETTER GRADE

Assessment will be based on the following catalog

Assessment	Midterm	Final	Project	TOTAL
Points	30	45	25	100

Grades versus minimum total points earned from the exams:

Points	90	80	75	70	65	60	55	50	45	40
Grade	A	A-	B+	B	B-	C+	C	C-	D+	D

EXPECTED STUDENT SEMESTER WORKLOAD	
Activities Carried Out in The Presence of An Instructor	**Total Hours in Semester**
Lecture (Including Discussion Sessions)	42
Recitation/Practice	
Laboratory/Studio/Kitchen/Flight	
Field Work	
Activities Carried Out by The Learner Him/Herself	**Total Hours in Semester**
Working on Assignments	30
Pre-class Learning of Course Material	18
Review of Course Material for Exam Preparation	50
Other	40
Overall Total Hours in Semester	180

RULES FOR THE USE OF AI IN THE COURSE	
Use of AI in the Course **(Free Use within Ethical Framework)**	You may **freely** use AI tools (ChatGPT, Copilot, Gemini, Claude, Grok, DeepSeek, etc.) in this course, provided that you do so **ethically** and in compliance with the **Principles for the Responsible Use of AI** listed below. You may benefit from AI in assignments and other learning activities. However, you must clearly acknowledge AI contributions, act ethically and transparently, and comply with university policies.
Principles for the Responsible Use of AI	1. **Use AI as a tool to support your learning, not as the ultimate source of knowledge:** AI can assist your learning and work, but the final judgment, creativity, and ownership of the output must always remain with you. 2. **Be transparent about AI's contribution:** If you use AI to generate, summarize, or improve any part of your work, you must clearly acknowledge it. Transparency protects your academic honesty and strengthens professional integrity. 3. **Recognize the limitations and biases of AI:** AI may produce inaccurate, incomplete, or biased outputs. Always review the results critically and verify them before using. 4. **Consult diverse sources, not just AI:** Do not rely solely on AI for information. Support, compare, and cross-check AI outputs with other valid and reliable sources. 5. **Protect your data and privacy:** Never share personal, sensitive, or confidential information with AI tools. Safeguarding such information is your responsibility.

OTHER RULES AND INFORMATION	
Health	The mode of instruction is subject to change in case of pandemic or other similar unavoidable reasons in line with the directives of the authorized public bodies. In such a case, the necessary revisions will be binding for all members of the OzU community, and will be announced by the relevant units, in particular Student Services.
Safety	Announced safety procedures must be strictly followed during any laboratory and similar work. Please refer to the safety guidelines posted in the laboratories. In case of emergencies, call 9911 (216 564 9911).
Accessibility	Reasonable accommodations will be provided for students with verifiable disabilities. Please inform your instructor if you need any assistance.

Academic and Technical Support	Students can get help from the Writing Center, Solution Center and IT in case they need academic and/or technical support with their classes.
Objections to Examination Results	Students who object to their exam results and cannot resolve their objections with the instructor of the course, may file a written petition for a review of their exam papers at their Dean's/Director's Office within five business days from the announcement their exam results.
Academic Integrity and Honor Code	Students are assumed to abide by scholastic honesty and uphold the Honor Code. Scholastic dishonesty includes, but is not limited to cheating, attempting to cheat, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting the work of another person, or tampering with the academic work of other students. Any form of scholastic dishonesty is a serious academic violation and will result in a disciplinary action and also may result in student getting a 0 grade in the related assessment.
Unauthorized Recording/Sharing of Lectures	Recording/sharing of synchronous/asynchronous lectures or copying of lecture recordings are prohibited without the permission of the instructor. Actions such as unauthorized recording, giving the recording to someone else, taking it, using it or sharing it in physical or virtual media are prohibited and/or are offenses according to the relevant legislation. In such cases, legal procedures will be initiated in addition to starting disciplinary actions against those involved.
Flexibility	Circumstances may arise during the course that prevents the instructor from fulfilling each and every component of this syllabus; therefore, the syllabus is subject to change. Students will be notified prior to any changes.
Principles for Classroom Behavior	Creating a safe space where ideas can be freely discussed in our classroom is everyone's responsibility. In accordance with the relevant disciplinary regulations, any verbal or written expression that disrupts the order of the class, involves the unauthorized distribution of materials, or is discriminatory or violent, and that violates the dignity of individuals in the classroom (students or educators) is strictly prohibited. During class, all participants are expected to respect each other's rights.